

```
print("INITIAL STATE:")
display(initial)

print("GOAL STATE:")
display(goal)

solution = dfs(initial, goal)

if solution:

    print("\nDFS SOLUTION FOUND")
    print("Number of moves:", len(solution) - 1)

    print("\nSolution Path:")

    for step, state in enumerate(solution):

        print("\nStep", step)
        display(state)

else:
    print("No solution found.")
```

INITIAL STATE:

```
-----
| 1 | 2 | 3 |
| 4 | 0 | 6 |
| 7 | 5 | 8 |
-----
```

GOAL STATE:

```
-----
| 1 | 2 | 3 |
| 4 | 5 | 6 |
| 7 | 8 | 0 |
-----
```

DFS SOLUTION FOUND
Number of moves: 434

Solution Path:



```
print("INITIAL STATE:")
display(initial)

print("GOAL STATE:")
display(goal)

solution = dfs(initial, goal)

if solution:

    print("\nDFS SOLUTION FOUND")
    print("Number of moves:", len(solution) - 1)

    print("\nSolution Path:")

    for step, state in enumerate(solution):

        print("\nStep", step)
        display(state)

else:
    print("No solution found.")
```

```
| 1 | 2 | 3 |
| 4 | 5 | 6 |
| 0 | 7 | 8 |
-----
```

Step 433

```
| 1 | 2 | 3 |
| 4 | 5 | 6 |
| 7 | 0 | 8 |
-----
```

Step 434

```
| 1 | 2 | 3 |
| 4 | 5 | 6 |
| 7 | 8 | 0 |
-----
```

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Code

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JupyterLab

Python 3 (ipykernel)

display(state)

INITIAL STATE:
(1, 2, 3)
(4, 0, 6)
(7, 5, 8)

GOAL STATE:
(1, 2, 3)
(4, 5, 6)
(7, 8, 0)

=====

DEPTH LIMIT: 0
=====

Depth 0
(1, 2, 3)
(4, 0, 6)
(7, 5, 8)

=====

DEPTH LIMIT: 1
=====

Depth 0
(1, 2, 3)
(4, 0, 6)
(7, 5, 8)

Depth 1
(1, 0, 3)
(4, 2, 6)
(7, 5, 8)

(1, 2, 3)
(4, 5, 6)
(7, 0, 8)

(1, 2, 3)
(0, 4, 6)
(7, 5, 8)

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Code

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JupyterLab

Python 3 (ipykernel)

(1, 2, 3)

(4, 5, 6)

(7, 0, 8)

(1, 2, 3)

(0, 4, 6)

(7, 5, 8)

(1, 2, 3)

(4, 6, 0)

(7, 5, 8)

=====

DEPTH LIMIT: 2

=====

Depth 0

(1, 2, 3)

(4, 0, 6)

(7, 5, 8)

Depth 1

(1, 0, 3)

(4, 2, 6)

(7, 5, 8)

(1, 2, 3)

(4, 5, 6)

(7, 0, 8)

Depth 2

(0, 1, 3)

(4, 2, 6)

(7, 5, 8)

(1, 3, 0)

(4, 2, 6)

(7, 5, 8)

(1, 2, 3)

(4, 5, 6)

(7, 0, 8)

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JupyterLab Python 3 (ipykernel)

```
Depth 2
(0, 1, 3)
(4, 2, 6)
(7, 5, 8)

(1, 3, 0)
(4, 2, 6)
(7, 5, 8)

(1, 2, 3)
(4, 5, 6)
(0, 7, 8)

(1, 2, 3)
(4, 5, 6)
(7, 8, 0)

GOAL FOUND AT DEPTH: 2

===== FINAL SOLUTION =====

Step 0
(1, 2, 3)
(4, 0, 6)
(7, 5, 8)

Step 1
(1, 2, 3)
(4, 5, 6)
(7, 0, 8)

Step 2
(1, 2, 3)
(4, 5, 6)
(7, 8, 0)
```

[]: